

SAFETY DATA SHEET

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

1.1 Product Details

Name : Oxygen, Compressed
Use of substance : Industrial and Specialty Gas Applications

1.2 Company Identification

Manufacturer's Name : Leeden Gases Sdn. Bhd
Manufacturer's Address : Lot PT 5074&5075, Jalan Jangur 28/43,
Seksyen 28, 40400 Shah Alam, Selangor.
Emergency telephone Number : 03-55228222 (Hunting Line) / 03-55228288 (Hotline)

SECTION 2: HAZARD IDENTIFICATION

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
Classification of the substance : OXIDIZING GASES - Category 1.
GASES UNDER PRESSURE - Compressed gas.
Hazard pictograms :



GHS03 GHS04

Signal word : DANGER
Hazard statements : May cause or intensify fire; oxidizer.
Contains gas under pressure; may explode if heated.

Precautionary statements:

Prevention : Keep away from clothing, incompatible materials and combustible materials.
Keep reduction valves free from grease and oil.
Response : In case of fire: Stop leak if safe to do so.
Storage : Protect from sunlight when ambient temperature exceeds 52°C/125°F. Store in a well-ventilated place.
Disposal : Dispose in accordance with all applicable regulations.
Other : Rapid release of compressed gas may cause frostbite.
Hazards

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Chemical name : Oxygen
CAS number : 7782-44-7
Proportion : 100%

SECTION 4: FIRST AID MEASURES

Inhalation : If adverse effects occur, remove to uncontaminated area. Get medical attention.

Skin contact : If frostbite or freezing occur, immediately flush with plenty of lukewarm water (105-115°F; 41-46°C). DO NOT USE HOT WATER. If warm water is not available, gently wrap affected parts in blankets. Get immediate medical attention.

Eye contact : Flush eyes with plenty of water for at least 15 minutes. Then get immediate medical attention.

Ingestion : If swallowed, get medical attention.

Most Important Symptoms/Effects:

Acute : Frostbite

Delayed : No information on significant adverse effects.

Note to Physicians : Treat symptomatically and supportively.

SECTION 5: FIRE FIGHTING MEASURES

Suitable extinguishing media : Carbon dioxide, regular dry chemical, Large fires: Use water spray, fog or regular foam.

Specific hazards arising from the chemical : Oxidizer. May ignite or explode on contact with combustible materials. Containers may rupture or explode if exposed to heat.

Hazardous Combustion Products : miscellaneous decomposition products

Fire Fighting Measures : Move container from fire area if it can be done without risk. Do not direct water at source of leak or safety devices; icing may occur. Cool containers with water spray until well after the fire is out. Stay away from the ends of tanks. For fires in cargo or storage area: Cool containers with water from unmanned hose holder or monitor nozzles until well after fire is out. If this is impossible then take the following precautions: Keep unnecessary people away, isolate hazard area and deny entry. Let the fire burn. Use extinguishing agents appropriate for surrounding fire. Cool

containers with water. Apply water from a protected location or from a safe distance.

Special Protective Equipment and Precautions for Firefighters : Wear full protective firefighting gear including self-contained breathing apparatus (SCBA) for protection against possible exposure.

SECTION 6: ACCIDENTAL RELEASE MEASURE

Personal precautions, protective equipment and emergency procedures : Wear personal protective clothing and equipment, see Section 8.

Methods and materials for containment and cleaning up : Avoid contact with combustible materials. Do not touch or walk through spilled material. Stop leak if possible without personal risk. Do not direct water at spill or source of leak. Keep unnecessary people away, isolate hazard area and deny entry. Isolate area until gas has dispersed. Ventilate closed spaces before entering.

Environmental precautions : Avoid release to the environment.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling : Keep away from clothing and other combustible materials. Keep reduction valves free from grease and oil. Keep away from heat/sparks/open flames/hot surfaces. No smoking. Keep container tightly closed. Wash hands thoroughly after handling. Do not eat, drink or smoke when using this product. Damaged cylinders should be handled only by specialists. Open valve slowly. Use only with equipment cleaned for oxygen service. To avoid fire or explosion, ground and bond container and receiving equipment (and ground personnel) before transferring material.

Conditions for Safe Storage, Including any Incompatibilities : Protect from sunlight. Store in a well-ventilated place. Store in accordance with all current regulations and standards. Protect from physical damage. Avoid heat, flames, sparks and other sources of ignition. Store in a clean, cool, dry place. Store below 52 C. Subject to storage regulations: U.S. OSHA 29 CFR 1910.101. Keep separated from incompatible substances.

Incompatible Materials : combustible materials, halo carbons, metals, bases, reducing agents, amines, metal salts, oxidizing materials

SECTION 8: EXPOSURE CONTROL AND PERSONAL PROTECTION

- Component Exposure Limits : The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit. At this time, the other constituents have no known exposure limits.
- EU - Occupational Exposure (98/24/EC) - Binding : There are no biological limit values for any of this product's components.
- Biological Limit Values and Health Surveillance Measures
- ACGIH - Threshold Limit Values - Biological Exposure Indices (BEI) : There are no biological limit values for any of this product's components.
- Engineering Controls : Provide local exhaust ventilation system. Ensure compliance with applicable exposure limits. Use a back flow preventive device in the piping. Use only equipment of compatible materials of construction. Use only with equipment rated for cylinder pressure. Store in receptacles with relief valves, grounding and bonding, and secondary containment.

Individual Protection Measures, such as Personal Protective Equipment

- Eye/face protection : Wear splash resistant safety goggles. Contact lenses should not be worn. Provide an emergency eye wash fountain and quick drench shower in the immediate work area.
- Skin Protection : For the gas: Protective clothing is not required. For the liquid: Wear appropriate protective, cold insulating clothing.
- Respiratory Protection : Any self-contained breathing apparatus that has a full face piece and is operated in a pressure-demand or other positive-pressure mode.
- Glove Recommendations : Wear insulated gloves.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

- Physical state : Gas.
- Colour : Colourless
- Odor : Odorless
- Molecular weight : 31.9988 g/mol
- Molecular formula : O₂
- Boiling/condensation point : -182.96 °C (-297 °F)
- Melting/freezing point : -218.4 °C (-361 °F)
- Flammability : Non-flammable
- Vapour Density (air=1) : 1.43
- Specific Gravity (water=1) : 1.14 at -183 °C

Vapour pressure	: 760 mmHg @ -183 °C
Water Solubility	: 3.2 % (@ 25 °C)
Viscosity	: 0.02075 cp
Density	: 1.309 g/L at 25 °C
Oxidising properties	: Oxidizer
Solvent Solubility	: Soluble in alcohol

SECTION 10: STABILITY AND REACTIVITY

Reactivity	: No reactivity hazard is expected.
Chemical stability	: Stable at normal temperatures and pressure.
Possibility of hazardous reactions	: Will not polymerize. Oxidizer
Condition to avoid	: Avoid contact with combustible materials. Protect from physical damage and heat. Containers may rupture or explode if exposed to heat.
Incompatibles	: Combustible materials, halo carbons, metals, bases, reducing agents, amines, metal salts, oxidizing materials
Hazardous decomposition products	: Miscellaneous decomposition products

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Likely Routes of Exposure

Inhalation	: Irritation, changes in body temperature, nausea, difficulty breathing, irregular heartbeat, chest pain, lung damage, dizziness, Disorientation, hallucinations, mood swings, pain in extremities, tremors, lung congestion, convulsions, cough
Skin Contact	: Frostbite, blisters
Eye Contact	: Irritation, blurred vision
Ingestion	: Ingestion of a gas is unlikely
Immediate Effects	: Frostbite

SECTION 12: ECOLOGICAL INFORMATION

No known ecological damage caused by this product.

SECTION 13: DISPOSAL INFORMATION

Disposal Methods:

Dispose of contents/container in accordance with local/regional/national/international regulations.

SECTION 14: TRANSPORT INFORMATION

UN Number : UN1072
Hazard class / Division : 2.2 (Non-flammable gas)



Subsidiary Risk : 5.1 (Oxidizing agent)



SECTION 15: REGULATORY INFORMATION

National Regulations : Occupational Safety and Health Act 1994 and relevant regulations:

- Occupational Safety and Health (Classification, Labeling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013.
- Occupational Safety and Health (Use And Standards of Exposure of Chemicals Hazardous to Health) Regulations 2000

Environment Quality Act 1974 & regulations:

- Environment Quality (Clean Air) Regulations 2014
- Environmental Quality (Scheduled Wastes) Regulations 2005

SECTION 16: OTHER INFORMATION

NFPA ratings (scale 0 - 4)

